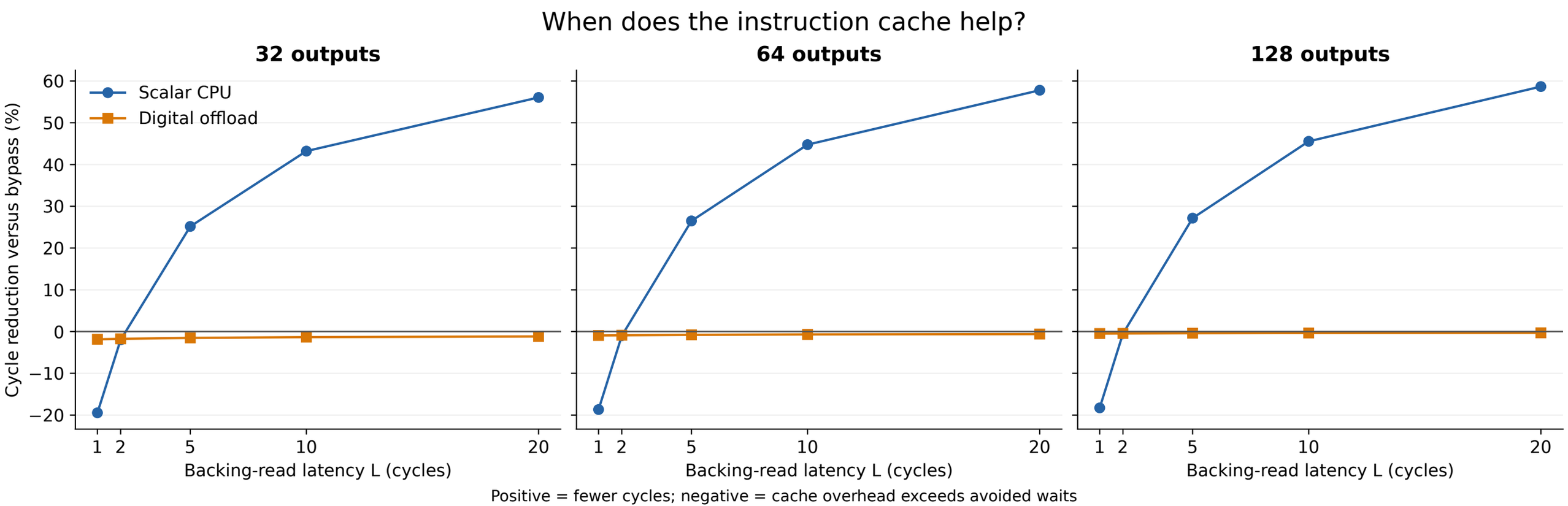
When does the instruction cache help?
32 outputs
64 outputs
128 outputs
Scalar CPU
Digital offload
Cycle reduction versus bypass (%)
60
50
40
30
20
10
0
−10
−20
1
2
5
10
20
Backing-read latency L (cycles)
Positive = fewer cycles; negative = cache overhead exceeds avoided waits